

Consider the following quadratic function:

$$f(x_1, x_2) = 2x_1^2 + 9x_2^2. \quad (1)$$

1. Calculate the condition number of the Hessian of  $f$  at point  $(x_1, x_2)$ .
2. Apply the change of variable  $x' = Mx$  where

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 3\sqrt{2} \end{pmatrix},$$

that is,

$$\begin{aligned} x'_1 &= 2x_1, \\ x'_2 &= 3\sqrt{2}x_2. \end{aligned}$$

Consider the function

$$\tilde{f}(x') = f(M^{-1}x').$$

Calculate the condition number of the Hessian (using the 2-norm) of  $\tilde{f}$  at point  $(x'_1, x'_2)$ .

Remember that, if  $A \in \mathbb{R}^{n \times n}$  is a non singular symmetric matrix, then the condition number of  $A$  is

$$\kappa(A) = \|A\| \|A^{-1}\|. \quad (2)$$

The norm 2 of a matrix is its largest singular value. Therefore, in this case,

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}, \quad (3)$$

where  $\sigma_1$  is the largest singular value of  $A$  and  $\sigma_n$  is the smallest. By extension, the condition number of a singular matrix (i.e., such that  $\sigma_n = 0$ ) is  $+\infty$ . If  $A$  is symmetric positive semidefinite, the singular values of  $A$  are its eigenvalues.